Sahaj Bhadja

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Summary

Software Developer with 2+ years building **Python, Linux, high-concurrency applications**, scalable data pipelines, automated deployments, and observable cloud services using networking fundamentals, clean OOD, and rigorous validation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Computer Networks, Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked asynchronous FastAPI orchestration with explicit job states and failure isolation, processing 1K+ concurrent AI jobs at 3x throughput while sustaining **high-concurrency reliability, monitoring, and clean modular design**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **performance analysis, data normalization, query tuning, and production troubleshooting**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **asynchronous execution, testing, debugging, and validation**.
- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **REST APIs, Git collaboration, and iterative releases**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch Decision Transformer for autonomous-driving reinforcement learning, improving planning 18% while processing 100K+ trajectories through reproducible **Linux evaluation pipelines, detailed performance benchmarking, and systematic research experiments**.
- Designed spatial-temporal encoders combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% through **geographic data processing, numerical analysis, and documented validation**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform services and scalable data pipelines using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across distributed production systems.
- Automated 20+ unit and integration suites with PyTest, Mockito, Jenkins, and virtual machines across Windows, Linux, and macOS, strengthening **CI/CD, Git workflows, release validation, system reliability, and maintainability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, object-oriented design, and event-driven communication** across distributed business locations.
- Automated Docker and Jenkins delivery, reducing deployment time 50% while using Sentry, Azure Monitor, and Kibana for **infrastructure monitoring, issue diagnosis, and reliable production releases** across environments.

Skills

Languages: Python, Go, Bash, C, C++, Java, JavaScript, TypeScript, SQL, Shell Scripting
Networking: TCP/IP, UDP, IP Addressing, Routing, BGP, DNS, HTTP, Network Probes, Traffic Measurement, Anycast
Systems & Data: Linux, Concurrency, Multithreading, OOD, Algorithms, MySQL, PostgreSQL, Redis, Data Normalization, Enrichment
Cloud & DevOps: Ansible, Docker, Terraform, Kubernetes, AWS, GCP, Azure, Jenkins, GitHub Actions, CI/CD, Git
Quality: Unit Testing, Integration Testing, PyTest, Debugging, Performance Profiling, Monitoring, Documentation, Code Reviews, Agile

Projects

GeoProbe Engine | *Python, Go, Linux, BGP, DNS, MySQL, Docker*

- Built a **high-concurrency IP geolocation pipeline** that runs distributed latency probes, resolves DNS, enriches BGP prefixes, normalizes geographic records, and persists confidence-scored results in MySQL for policy queries.

NetAtlas Automation | *Python, Bash, Ansible, Terraform, AWS, Prometheus*

- Designed a **network measurement and deployment toolkit** using Python probes, Bash diagnostics, Ansible configuration, Terraform infrastructure, and Prometheus metrics to validate routing behavior, detect anomalies, and automate recovery.

CPU Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, round-robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across interactive workloads through a browser-based scheduling simulator.